

AIM-5014: Ai Safety & Security (3 credits)

Fall 2026

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Class Hours: Wednesday 3:15–5:15pm
Classroom: 205 Lexington Avenue 401

Assignment 1: Due Wednesday, Sept. 16th

In a 3–5 page essay, trace how the Transformer architecture fundamentally resolved the computational and representational bottlenecks of classical sequence models. In your analysis, address the following core areas:

- **Gradient Dynamics & Bottlenecks:** Explain how recurrent architectures suffer from vanishing gradients when processing long contexts, and analyze how residual stream connections and layer normalization mathematically facilitate stable gradient backpropagation across deep stacks
- **Dynamic Contextualization:** Walk through the scaled dot-product self-attention mechanism—specifically defining the roles of Queries, Keys, and Values—and explain why this mechanism enables contextualization.
- **Sequence-to-Sequence vs. Causal Autoregression:** Compare the information flow in the original machine translation encoder-decoder (specifically cross-attention) with modern decoder-only causal language models (GPT-style) Detail how causal masking and cross-entropy loss against one-hot targets drive autoregressive next-token prediction.